Niranjan Kukkala

Innovation Engineer

+32 456 22 15 13

medium.com/@niru5

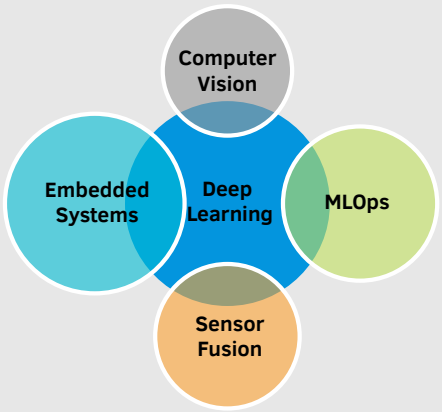
niranjan.reddy.wo@gmail.com

/in/niranjan5/

niru-5

Technical Skills

Overview



Programming

$0\ hours \longrightarrow 10000\ hours$

Python

C • C++

MATLAB

Node.js

Areas

Deep Learning | Machine Vision | Sensor Calibration | Embedded Programming | AWS Cloud | Model Quantization | MLOps | Data Pipelines

Tech Stack

Libraries

PyTorch | matplotlib | OpenCV | scikit-learn | MLflow

Boards

ESP32 | RPi | TDA4VM | Jetson TX2

Version Control

Git | Perforce | Subversion

Experience

Sep 2022 - Present — **Innovation Engineer** — CNH Industrial

- Building end-to-end computer vision pipelines: data collection, sensor placement, sensor selection, model training, model optimization, and deployment (kernel implementation) on edge devices (TI)
- Training foundation models on internal datasets, reducing annotated data needs by 70% for downstream applications
- Proficient in quantization techniques for deep learning models, applying quantization-aware training to run models at higher speeds on edge devices
- Writing IP disclosures (3 patents) and monitoring competitor IP
- Guiding Masters and Bachelors students (4 total) through thesis work, in collaboration with CNH and partner universities
- Handling the training infrastructure, building better methods for sharing the local compute cluster across teams and thesis students
- Writing specific kernels (application level) for the DSP (Digital Signal Processor) on the hardware
- Building algorithms for end-of-line testing of the hardware

Aug 2021 - Sep 2022 — **Machine Learning Engineer** — Arcadis

- Designed experiments for data collection
- Trained segmentation models for corn kernels
- Matured the edge deployment pipeline; deployed to Texas Instrument boards
- Worked as a consultant to CNH, and later joined CNH directly

Aug 2018 - Apr 2020 — **Associate Software Engineer** — Mathworks

- Implemented a Multi-Object Lidar Tracker and integrated it into a Lidar annotation tool
- Researched Video Object Segmentation and implemented “Fast Video Object Segmentation by Reference-Guided Mask Propagation” in MATLAB
- Surveyed state-of-the-art object tracking papers and implemented DASiamRPN (winner of the 2018 VOT challenge) in MATLAB
- Built a Lidar annotation app using an MVC framework

Open Source

May 2020 - Jul 2020 — **imusensor** — 100+ stars on GitHub

- Maintainer of imusensor (PyPI), a Python library for interfacing the MPU9250 (IMU) with Raspberry Pi, including calibration of gyroscope, accelerometer, and magnetometer
- Implemented Kalman and Madgwick filters for raw IMU values
- Documented the work with blog posts explaining the math behind calibration and filters

Education

Sept 2020 - June 2021 — **Masters of Artificial Intelligence** — KU Leuven

- Cum laude

Aug 2014 - Aug 2018 — **BTech., Electrical Engineering & Engineering Science** — IIT Hyderabad

- Minor: Entrepreneurship
- Grade: 8/10